

TANUSH THAKRAN

AI / ML Engineer · Applied Machine Learning · LLM & Retrieval Systems

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Portfolio • LinkedIn • GitHub

PROFILE

Machine learning engineer who ships models with evidence behind them — controlled ablations, leakage-safe evaluation, and metrics reported against honest baselines. Experience spans supervised learning-to-rank on multi-million-row datasets, deep learning with PyTorch and TensorFlow, and production retrieval-augmented generation systems. Comfortable owning the full path from raw data extraction through feature engineering, training, evaluation, and deployment.

TECHNICAL SKILLS

Machine Learning: Learning-to-Rank (LambdaRank) · Gradient Boosting (LightGBM) · Behaviour Cloning · Feature Engineering · Ablation Design & Controlled Experiments · Model Evaluation (NDCG, top-k accuracy) · Leakage-Safe Validation

Deep Learning: PyTorch · TensorFlow · Neural Networks · Seq2Seq with Attention · CNN Training & Diagnostics

LLM & Retrieval: Retrieval-Augmented Generation (RAG) · Embeddings & Vector Search · Cosine Similarity Retrieval · Prompt Engineering · Whisper (Speech-to-Text) · Ollama (Local LLMs)

Data Engineering: Pandas · NumPy · scikit-learn · PyArrow / Parquet · Multiprocessing · joblib · Large-Scale ETL

Languages: Python · JavaScript (ES6+) · Java · Dart · SQL-style query logic

Web & Mobile: React.js · Node.js · Express.js · Tailwind CSS · Flutter

Infra & Tools: Git · GitHub · MongoDB · Firebase · REST APIs · Cursor · GitHub Copilot · Vercel · Netlify · Render

MACHINE LEARNING PROJECTS

Pokémon TCG Move Prediction — Behaviour Cloning via Learning-to-Rank

Python · LightGBM · PyArrow

github.com/Tanush1206/pokemon-tcg-move-prediction

- Framed move selection as learning-to-rank rather than classification — menus vary in length each turn and option value is relative — and trained an **LGBMRanker with lambdarank** over 3.35M rows and 83 features.
- Reached **57.4% top-1 accuracy against a 21.7% random baseline (2.65x lift)** on 50,649 held-out decisions; the true move ranks in the model's top 3 for 88% of cases, making it a strong shortlister for downstream search.
- Built a parallel extraction pipeline converting 4,604 raw episode game trees into 4.77M feature rows in ~75s across 31 workers, streaming to Parquet row groups to avoid a ~20 GB in-memory blowup.
- Recovered per-option card identity from board state by reverse-engineering undocumented index semantics and validating the hypothesis across 1,758 decisions — **the single largest accuracy gain in the project at +0.054**.
- Maintained a 7-run ablation log on a fixed test set with episode-level splits to prevent turn-to-turn leakage; used a volume-matched control to disprove an apparent data-quality effect that was in fact training volume, and selected hyperparameters on validation only.

LIGHTGBM · LEARNING-TO-RANK · FEATURE ENGINEERING · EXPERIMENT DESIGN · PARQUET · MULTIPROCESSING

RAG-Based AI — Offline Retrieval-Augmented QA Pipeline

Python · Ollama · Whisper

github.com/Tanush1206/rag_based_ai

- Built a fully offline RAG pipeline that ingests course videos and answers natural-language questions grounded strictly in retrieved transcript context — no external APIs, everything runs locally.
- Engineered the end-to-end flow: audio extraction (ffmpeg), timestamped transcription with Whisper, embedding generation via the local bge-m3 model, and top-k retrieval using cosine similarity; batched chunk requests to eliminate runner crashes at scale.
- Fed only the most relevant chunks to a local deepseek-r1 LLM through Ollama, constraining answers to a cited video, title, and timestamp for traceable, hallucination-resistant responses.

RAG · EMBEDDINGS · VECTOR RETRIEVAL · WHISPER · LOCAL LLM

PactPal — AI Legal Agent

Python · NLP · React

pactpal-frontend.onrender.com

- AI-powered platform that converts complex legal contracts into plain language using NLP, reducing reading time by ~30% for non-expert users.
- Designed for startups and individuals, enabling faster, informed decisions without specialist legal review.

NLP · TEXT ANALYTICS · PRODUCTION

EXPERIENCE

App Developer Intern — CYPARTA

Cairo, Egypt · Dec 2025 – Jan 2026

- Developed the Driver & Medic mobile application focused on high-criticality logistics and real-time coordination, with end-to-end ownership.
- Implemented full trip workflows — request acceptance, live status tracking, navigation integration — and engineered post-trip EMS reporting modules for data integrity.
- Worked with real-time GPS telemetry streams, handling ingestion, state updates, and consistency across concurrent users.

FLUTTER · FIREBASE · REAL-TIME DATA · API INTEGRATION

Junior Manager — AIESEC in Bengaluru

Oct 2025 – Mar 2026

- Managed operations for the Outgoing Global Volunteer department, tracking departmental KPIs to steer team priorities.
- Transitioned to Incoming Global Volunteer, overseeing project management and intern integration.
- Led cross-functional teams to hit targets and expand organizational reach across Bengaluru.

LEADERSHIP · OPERATIONS · KPI REPORTING

ADDITIONAL PROJECTS

ATW Mobile — Driver & medic coordination app with real-time GPS and full trip-lifecycle management; shipped for CYPARTA, Cairo. *Flutter · Firebase*

CLIQUE — Event tracking and centralization platform for university students. *React · Node.js*

AASRAH — Humanitarian platform connecting users with NGOs, with Google Maps API location tracking. *React · Node/Express*

EDUCATION

BITS Pilani

2024 – 2028 · India

Integrated B.Sc. in Computer Science — Coursework: Algorithms, Data Structures, Machine Learning, Deep Learning, Database Systems, System Design.

Columbia Foundation School

2022 – 2024

Class 11th & 12th — Science & Mathematics.

CERTIFICATIONS

Intro to Machine Learning — Kaggle (Jul 2026) · The Ultimate Job Ready Data Science Course — CodeWithHarry · AI Fluency: Framework & Foundations — Anthropic (Jul 2026) · Python Course — Tutedude (Apr 2026) · Learn JavaScript — Scrimba (Feb 2026)